QP CODE: 24900269

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTER MGU-UGP (HONOURS)
REGULAR EXAMINATION NOVEMBER 2024
First Semester
Multi-Disciplinary Course - MG1MDCZGY100
ORNAMENTAL FISH FARMING AND AQUARIUM KEEPING
(2024 Admission onwards)

Duration: 1 Hour Maximum Marks: 35

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Part A

Fill in the Blanks
Answer **All** Questions. Each question carries **1** mark

1. **Chironomous Larvae / Blood Worms / Red Worms** are reddish worms found in drainage canals, where there is a gentle flow of water. [K] [1]
2. **Rotifers** are the free-swimming minute organisms in marine water known as wheel animalcules. [K] [1]
3. In biological filters, living organisms like **bacteria** are used. [K] [4]
4. The scientific name of Guppy fish is ***Poecilia reticulata***.(**Liberal Valuation**) [K] [3]
5. To prevent the contamination during transport, **antibiotics / starvation / anesthetizing agents** are used to ensure the water free from pathogens.(**Liberal Valuation**) [K] [2]

(5 × 1 = 5 Marks)

Part B

Short Questions
Answer **5** questions. Each question carries **2** marks

6. **Write an account on infusoria preparation as fish feed.** [K] [1]
Infusoria are tiny living single-celled organisms that serve as essential food for newly hatched fish larvae. To prepare them, place banana peels, cabbage leaves, potato pieces, spinach, dried beans, aquatic plants, or skimmed milk in water in a shady spot and cover with a fine mesh to promote bacterial growth. These bacteria feed on decaying organic matter, and infusoria grow by consuming the bacteria .Three tablespoons of infusoria rich water is enough for a 20-litre tank with 50 fish larvae.
(**Any relevant Points**)

7. **List down commonly used raw materials for making artificial feeds**. ***(Any Four Points)*** [K] [1]
Cereal powder, Tapioca powder, Rice or Wheat bran, Egg white, Fish meal, Prawn meal, Clam meal, Soybean meal, Groundnut oil cake, Coconut oil cake, Vegetable oils and Fish oils are some of the raw materials used as sources of carbohydrate, proteins, fats, vitamins and minerals.

8. **List the essential equipment in a quarantine tank.** ***(Any Two points)*** [K] [2]
Isolation tank, Aerators, Nets, Filters

9. **What is the use of insulating box during the transportation of fishes?** [K] [2]
An insulating box is used to maintain a consistent and controlled water temperature during transport. It reduces stress during fish transportation.

10. **Categorize the diseases caused by microbes in aquarium fishes based on the nature of the microbes and give examples of diseases under each category.** ***(Liberal Valuation)*** [An] [3]

Microbe Type	**Disease**
Bacteria	Dropsy, Fin Rot, Tuberculosis, Columnaris (Cotton Wool Disease), *Aeromonas* Infection
Virus	Lymphocystis, Koi Herpes Virus (KHV), Spring Viremia of Carp (SVC)
Fungus	Saprolegniosis, Fungal Gill Disease, Mycotic Infection

11. **Explain the treatment measures of any one viral disease in aquarium fishes. (*Liberal Valuation*)** [U] [3]
Any one microbial diseases (Viral / Bacterial / Fungal disease) and its treatment measures

12. **Write any two uses of filters in an aquarium. (Any Two Points)** [U] [4]
Toxic waste removal
Maintain water quality
Establishing and maintaining a balanced aquatic ecosystem.
Chemical filters prevent discoloration, odours and pollutants.
Controls algal growth
Enhancing overall filtration and minimize manual maintenance.

(5 × 2 = 10 Marks)

Part C

Short Essays

Answer **5** questions

Each question carries **4** marks

13. **Compare the culture methods of *Artemia*, Tubifex worms, and *Chironomous* larvae as live feed.** ***(Liberal Valuation)*** [An] [1]

***Artemia* (Brine Shrimp):** Salt pans with high salinity or saline water jars, Saline water (1-5g/l), temperature 27-30°C, aeration, 24-26 hours for hatching, attracted to light source, siphoned, washed. Can be fed live or in freeze-dried form
Tubifex Worms: Drainage canals with gentle water flow, Mixture of pig/sheep manure and sand, 15–60 days for thick mass, removed with spade or net, washed thoroughly. Fed as live feed to fish
***Chironomous* Larvae (Blood Worms):** Drainage canals, ponds with decaying vegetation, manured soil in cement tanks, Organic manure, natural habitat attracting flies, 3 days after hatching, collected using a sieve. Directly supplied to bigger fishes; killed and used for smaller fishes and larvae.

14. **Describe the role of vitamins and minerals in fish feed formulation, explaining their functions and deficiency.** ***(Liberal Valuation)*** [U] [1]
Mention of any two vitamins and minerals used for fish feed formulation- 2 marks each

15. **Explain the importance of anaesthesia during the transportation of ornamental fishes**. [K] [2]
Importance: Stress Reduction: Calms the fish, lowering metabolic activity and oxygen demand. Prevention of Injury: Reduces erratic movement, avoiding damage to fins or scales. Improved Survival Rate: Decreases mortality during long-distance transportation.
(**Any Four Points)**

16. **What is Quarantine of fishes? Explain the importance of quarantine.*(Liberal Valuation)*** [A] [2]

Quarantine is the practice of isolating new or sick fish in a separate tank before introducing them to the main aquarium.
Importances:
Prevention of Disease Spread: Stops infectious pathogens from infecting healthy fish.
Observation Period: Allows monitoring for diseases, parasites, or behavioral abnormalities.
Treatment: Provides a controlled environment for administering medications. Quarantine typically lasts 2–4 weeks with regular water changes and observation.
Stress Reduction, Pathogen Control and Improved Survival Rate:
(Definition -1 mark, Any three points: 3 marks)

17. **Explain the breeding technique of goldfish. *(Liberal Valuation)*** [A] [3]

Gold fishes are egg scatterers with no parental care. Females lay 5,000–15,000 adhesive eggs. Males have white spots on operculum and fins and females have soft, swollen abdomens. When mature male releases milt. 50% of water in the breeding tanks are replaced daily, two weeks before breeding. Males and females for breeding purpose are generally placed in smaller tanks with aquatic plants or materials (e.g., nylon fibers) for egg adhesion. Males stimulate females during courtship to release eggs.Fertilized eggs are translucent; unfertilized eggs turn white. Eggs hatch in 48–72 hours. After eggs are hatched breeders are removed to prevent egg consumption. No feed are given for 3 days. After 3 days live feed like infusoria, brine shrimp or boiled egg yolk. Are given to the growing fish fry.
(Any relevant points- 4 marks)

18. **Illustrate the role of the nitrogen cycle in an aquarium**. (***Out of Syllabus - Give full marks if attempted)*** [A] [4]

The nitrogen cycle ensures the breakdown of harmful waste products in an aquarium.
Fish excreta and decaying organic matter produce ammonia (toxic). Beneficial bacteria (*Nitrosomonas*) convert ammonia into nitrites (less toxic). Another bacterial group (*Nitrobacter*) converts nitrites into nitrates (relatively harmless). Plants absorb nitrates synthesize aminoacids, organic compounds.
Significance: Maintains water quality, prevents ammonia toxicity and ensures a healthy environment for fish and plants.

19. **Analyze the importance of filtration systems in maintaining a healthy aquarium environment. (*Liberal valuation*)** [A] [4]

Filtration systems are critical for maintaining a healthy aquarium environment, ensuring the well-being of aquatic organisms, and promoting sustainable aquarium management. Filtration systems effectively remove physical, chemical, and biological contaminants, ensuring clear and clean water. This supports the stability of water parameters. Mechanical filtration reduces particulate matter, preventing gill damage and respiratory issues in fish. Biological filtration converts toxic substances like ammonia and nitrite into less harmful nitrate, reducing stress and promoting fish health. By eliminating debris and organic particles that can harbor harmful microorganisms, filtration systems lower the risk of disease outbreaks, ensuring the aquarium remains a safe environment for the fishes. Clean water conditions promote better feed conversion, faster growth rates, and improved overall health of aquatic species. Filtration systems reduces the need for frequent water changes and minimizes the environmental impact of waste disposal.
(Any relevant points- 4 marks)

(5 × 4 = 20 marks)

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